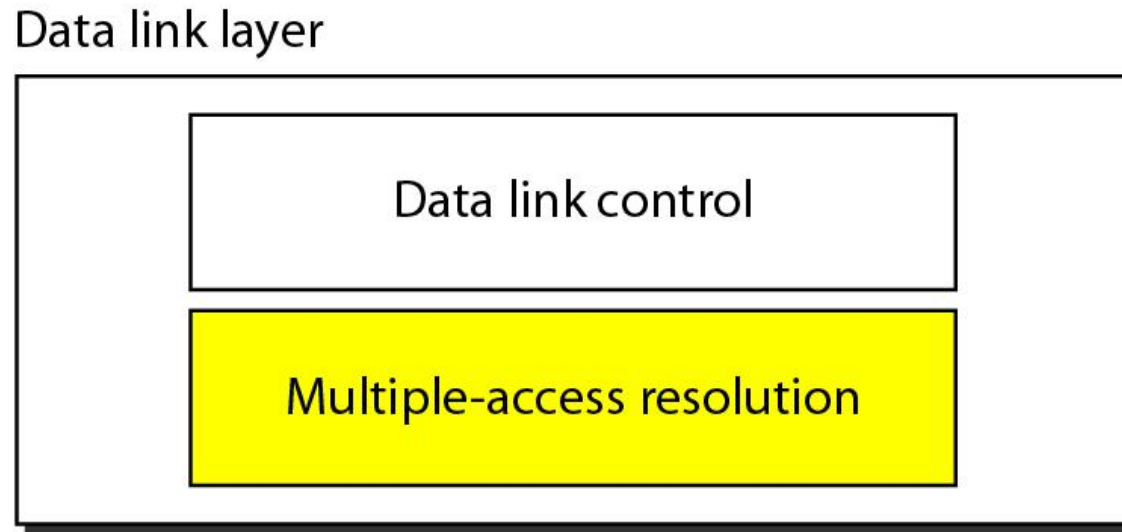


Media Access Control

Figure 12.1 *Data link layer divided into two functionality-oriented sublayers*



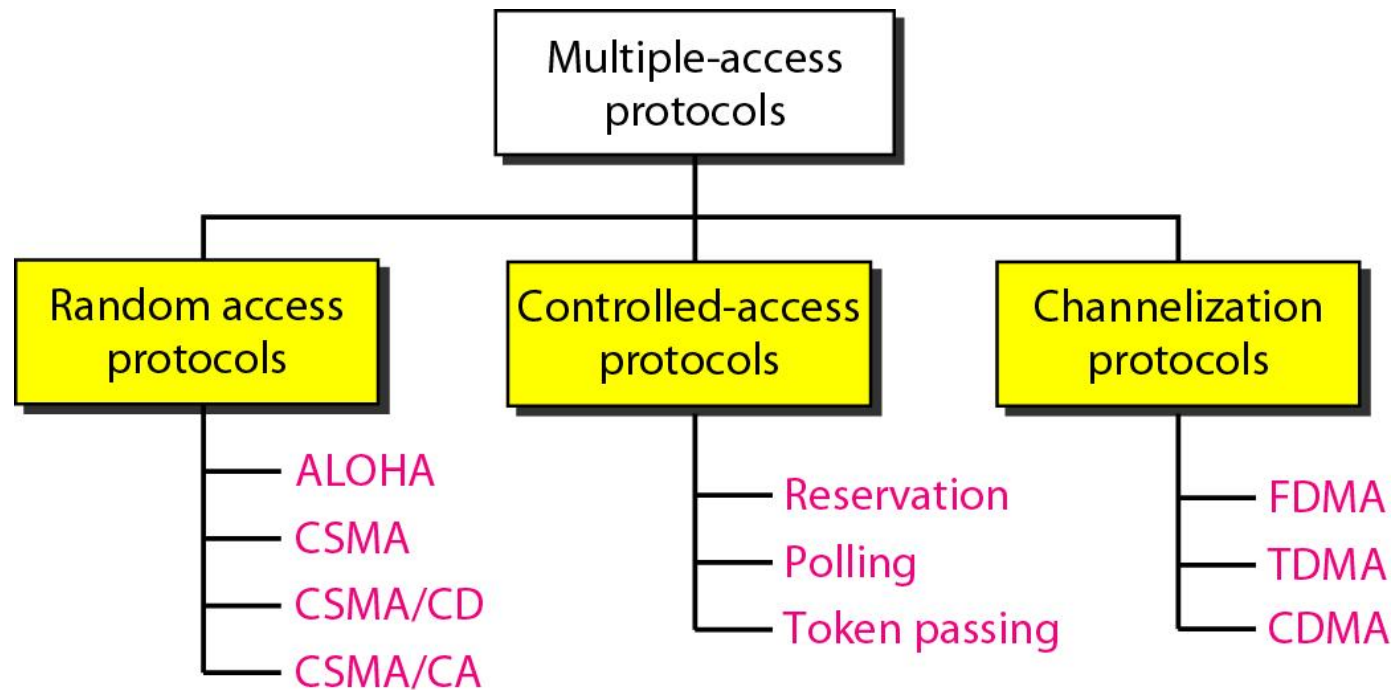
Upper sublayer – responsible for data link control

- **Called LLC – for flow and error control**

Lower sublayer - responsible for resolving access the shared media

- **Called MAC – for multiple access resolution**

Figure 12.2 *Taxonomy of multiple-access protocols discussed in this chapter*



12-1 RANDOM ACCESS

*In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. The decision depends on the state of the medium (idle or busy).*

Topics discussed in this section:

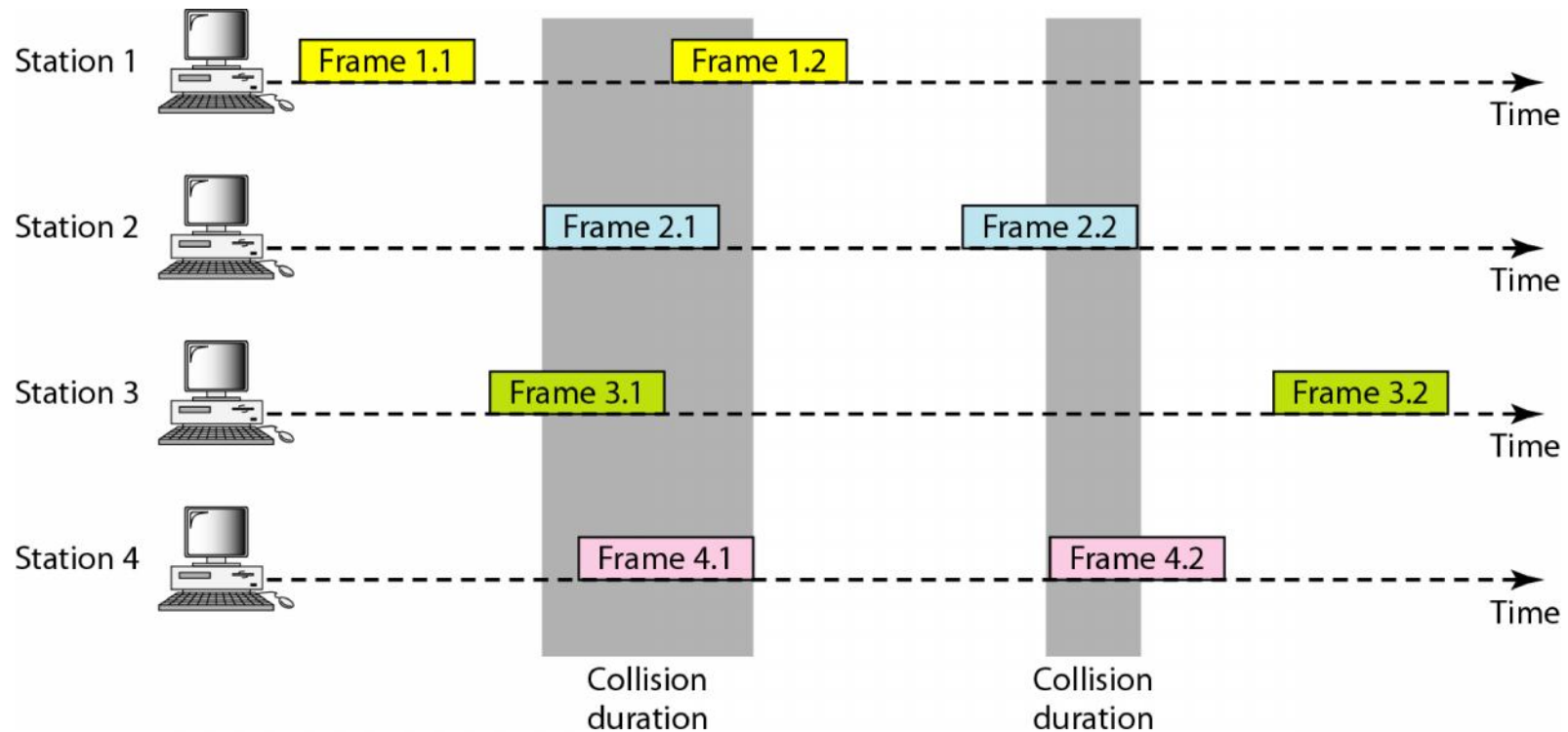
ALOHA

Carrier Sense Multiple Access

Carrier Sense Multiple Access with Collision Detection

Carrier Sense Multiple Access with Collision Avoidance

Figure 12.3 *Frames in a pure ALOHA network*



Pure ALOHA:

1. Each station sends a frame whenever it has a frame to send
2. One channel to share, possibility of collision between frames from different stations

Figure 12.4 *Procedure for pure ALOHA protocol*

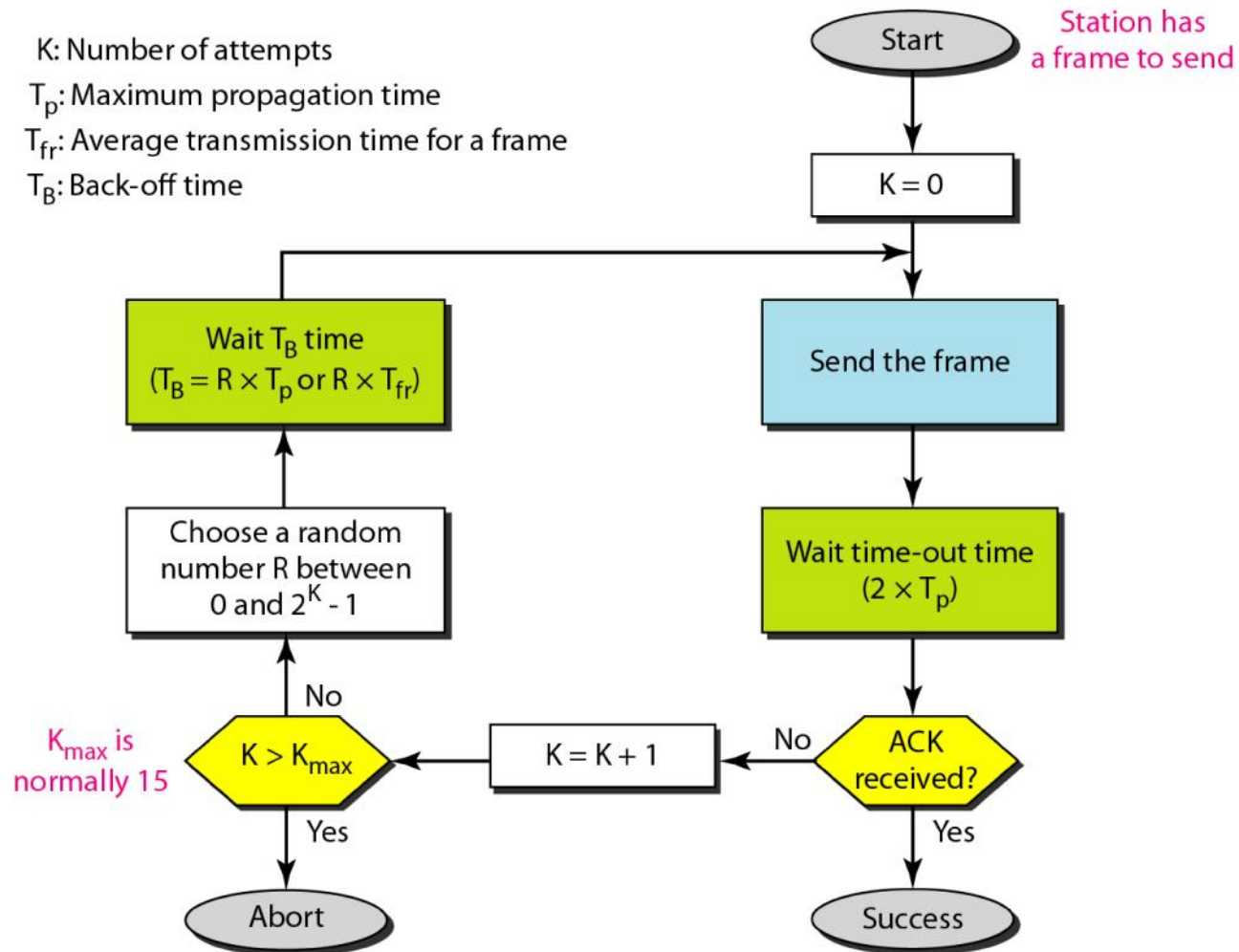


Figure 12.5 *Vulnerable time for pure ALOHA protocol*

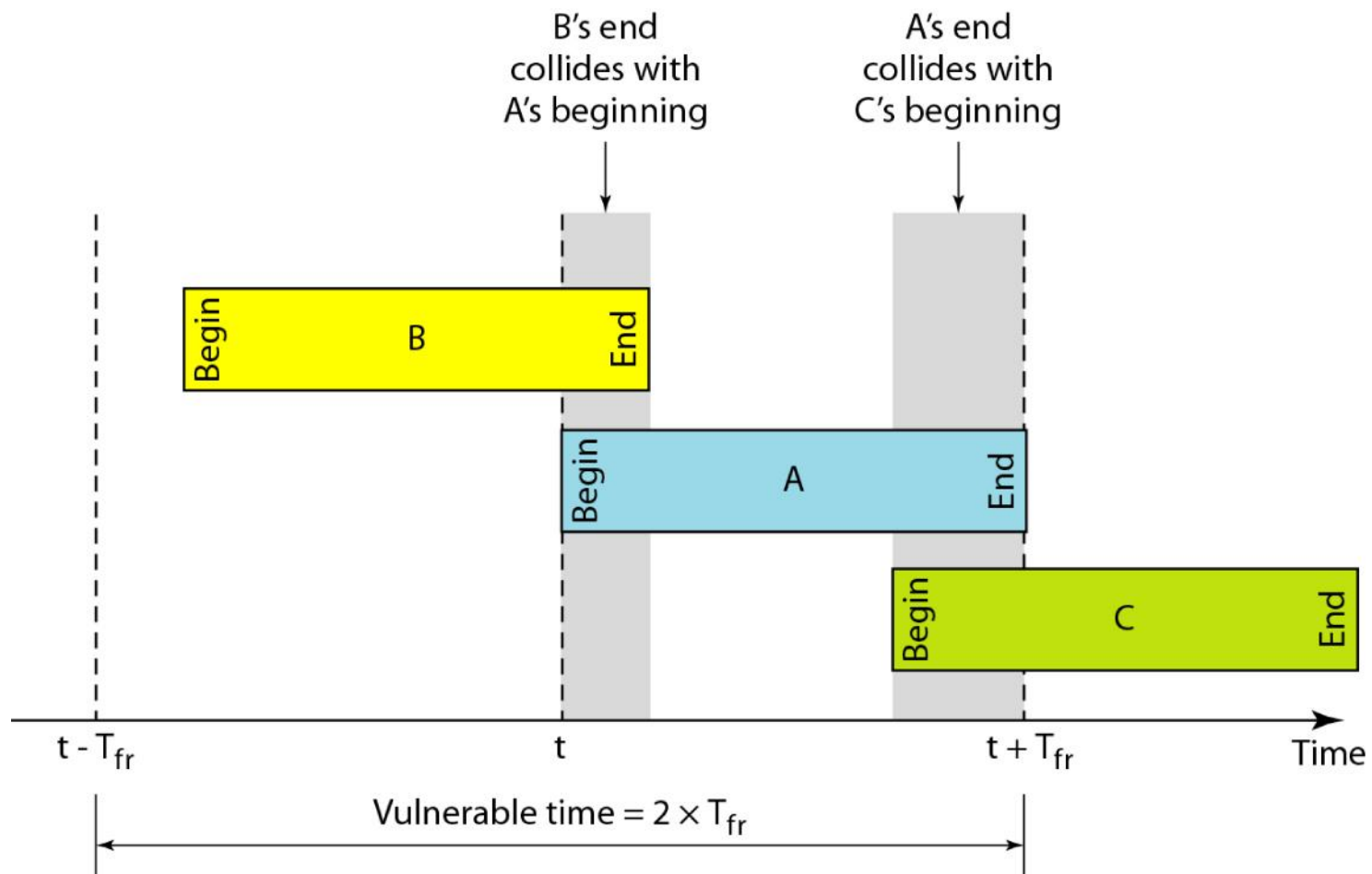
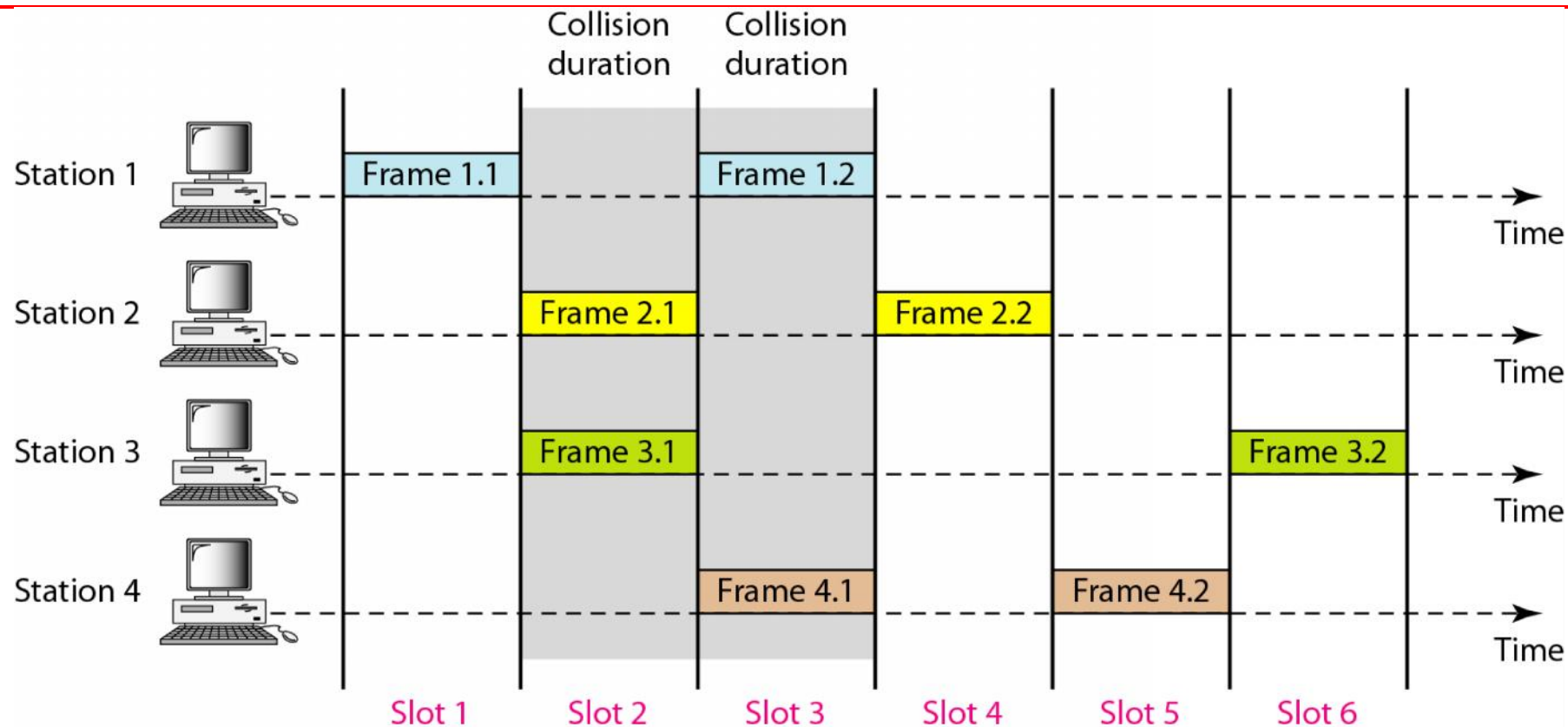


Figure 12.6 *Frames in a slotted ALOHA network*



Slotted ALOHA:

1. We divide the time into slots and force the station to send only at the beginning of the time slot

Figure 12.7 *Vulnerable time for slotted ALOHA protocol*

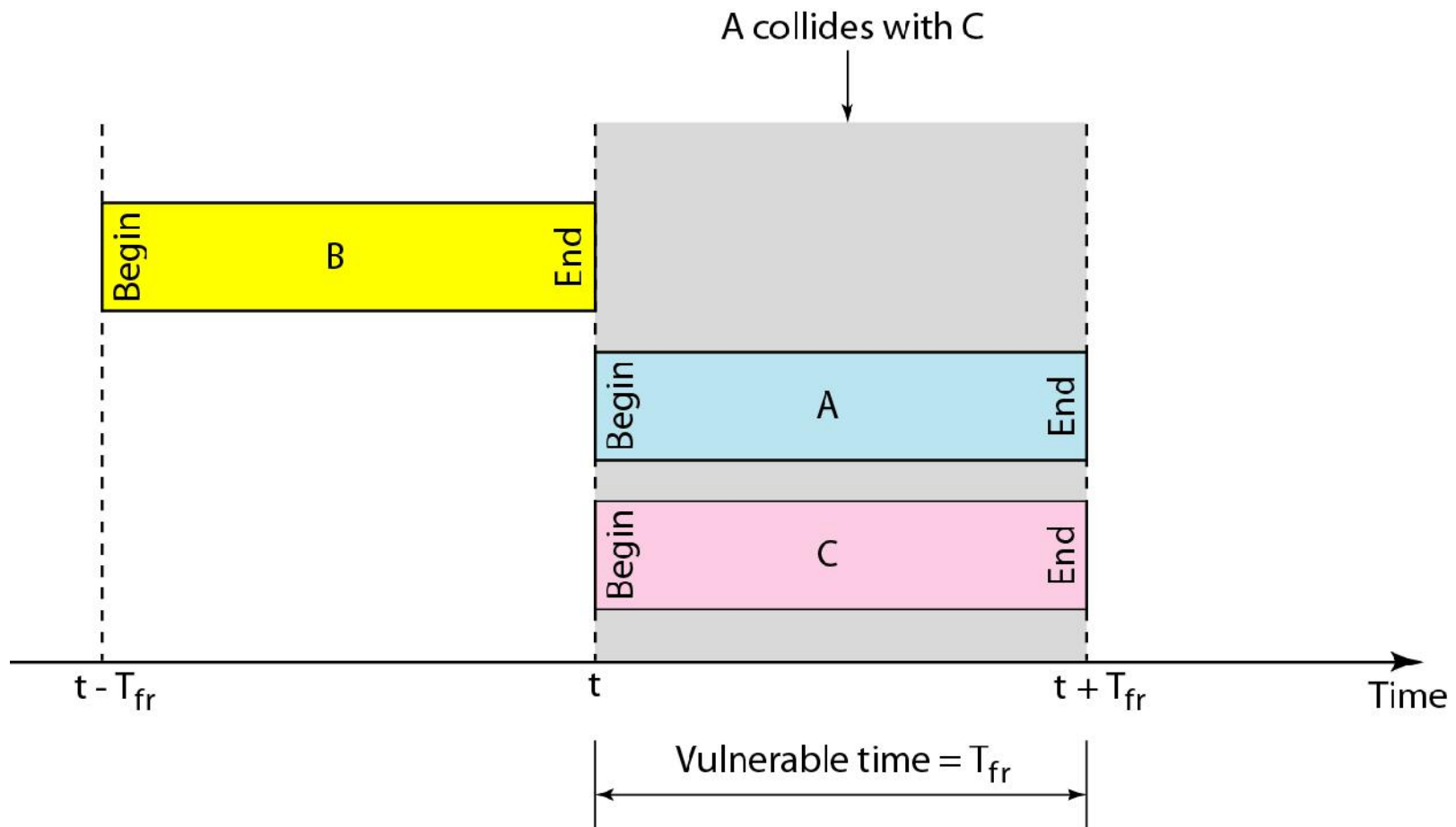
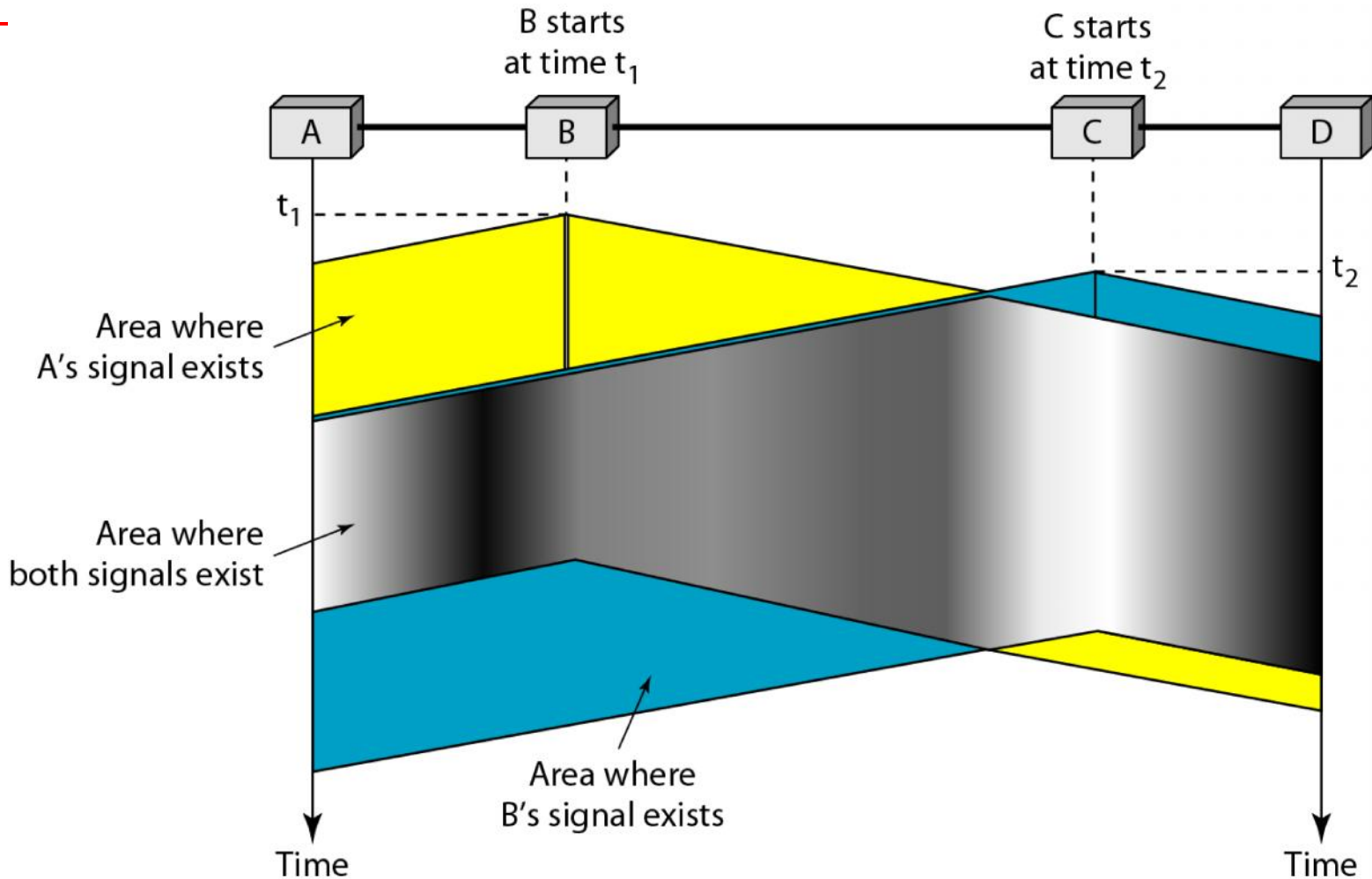


Figure 12.8 *Space/time model of the collision in CSMA*



CSMA – each station first listen to the medium before sending
Principle : “sense before transmit” or “listen before talk”

Figure 12.9 *Vulnerable time in CSMA*

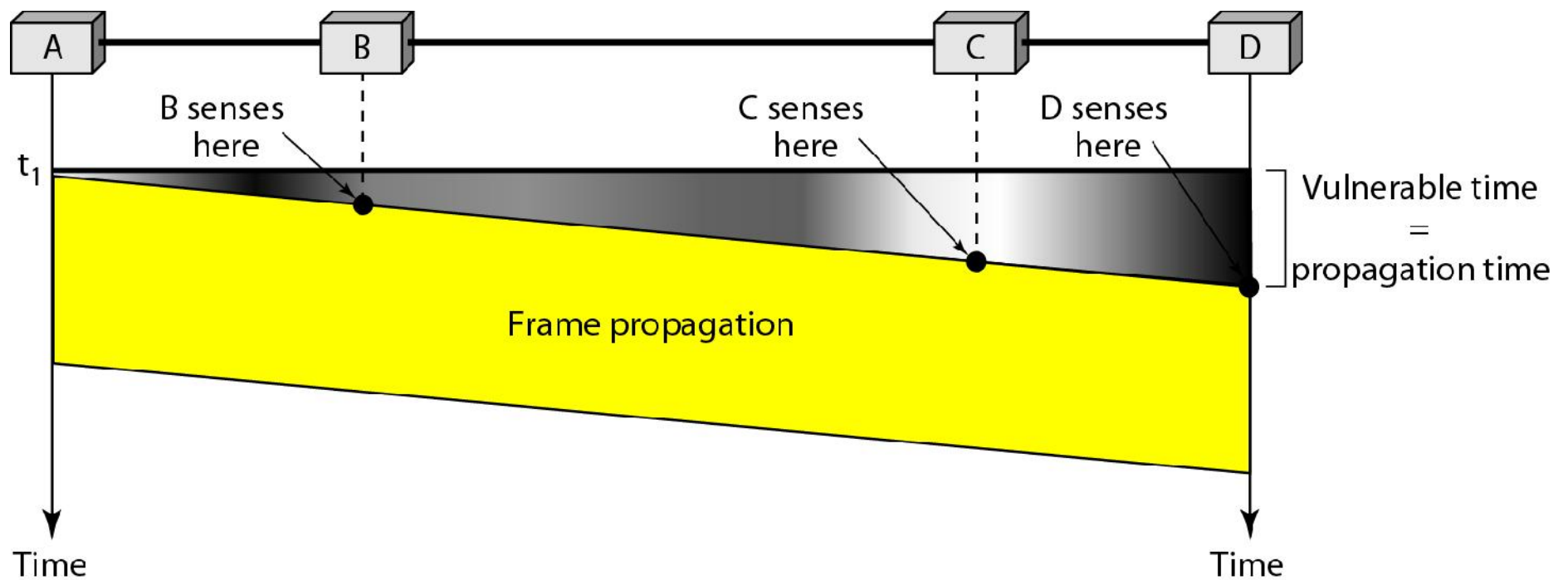
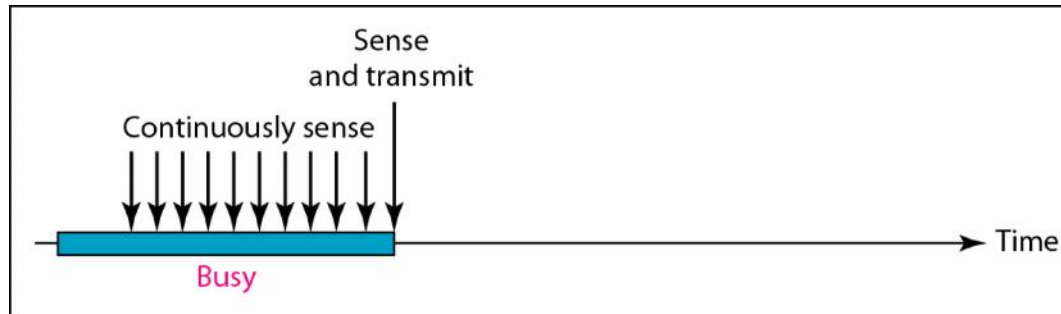
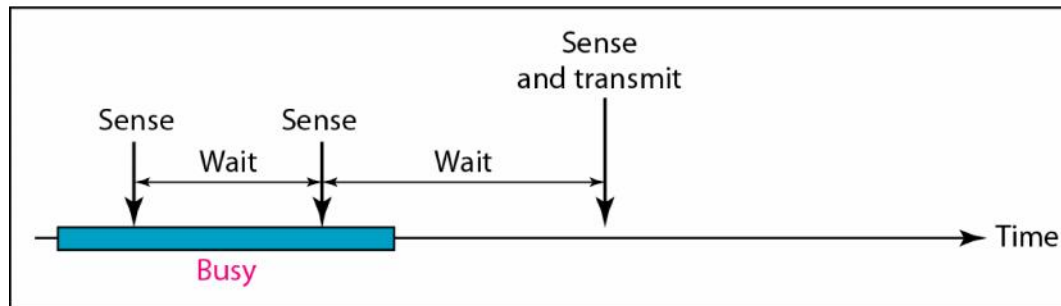


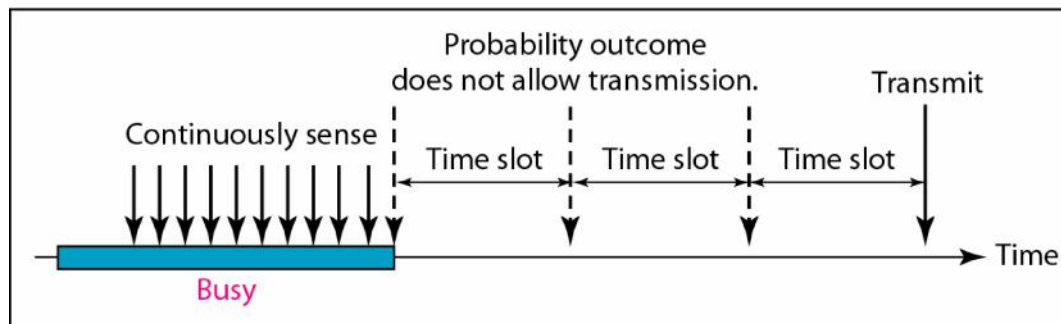
Figure 12.10 *Behavior of three persistence methods*



a. 1-persistent



b. Nonpersistent



c. p-persistent

1-Persistent-after station finds the line idle, send its frame

Nonpersistent-senses the line; idle: sends immediately; not idle: waits random amount of time and senses again

p-Persistent-the channel has time slots with duration equal to or greater than max propagation time

Figure 12.11 *Flow diagram for three persistence methods*

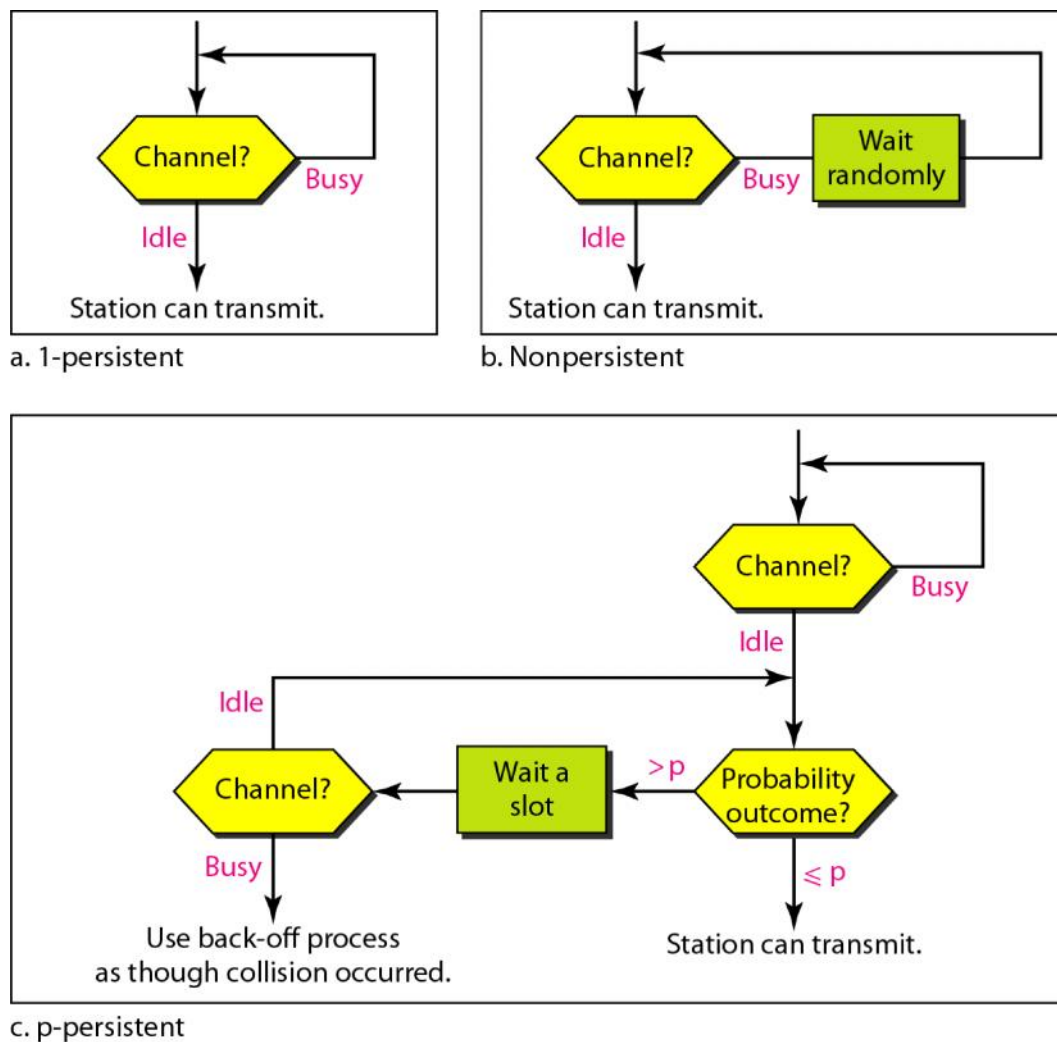


Figure 12.12 *Collision of the first bit in CSMA/CD*

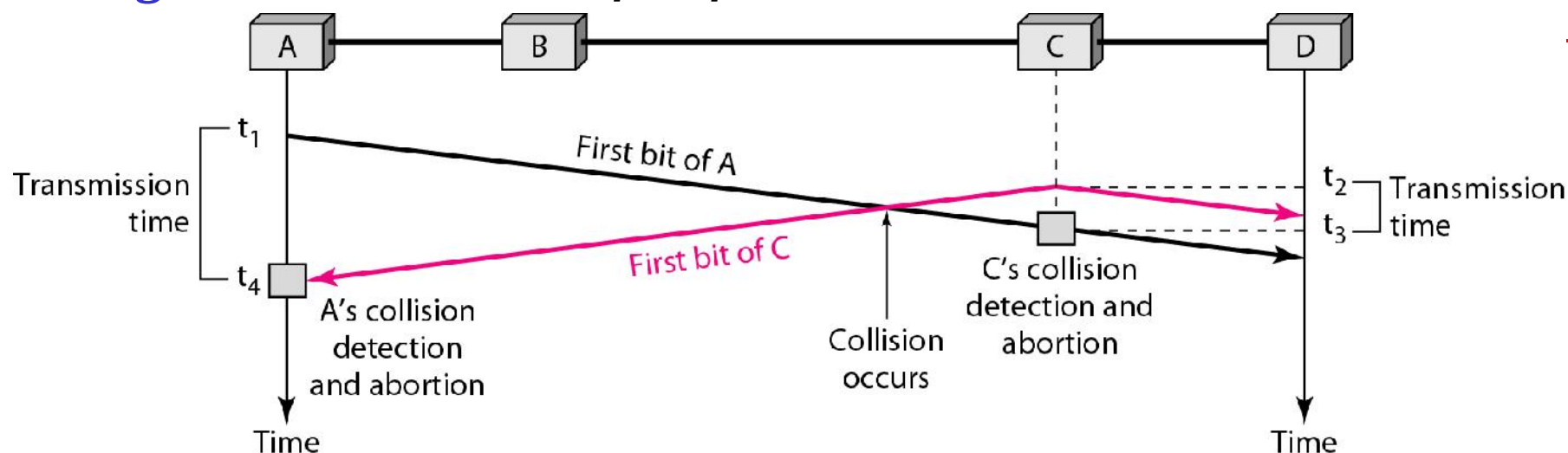
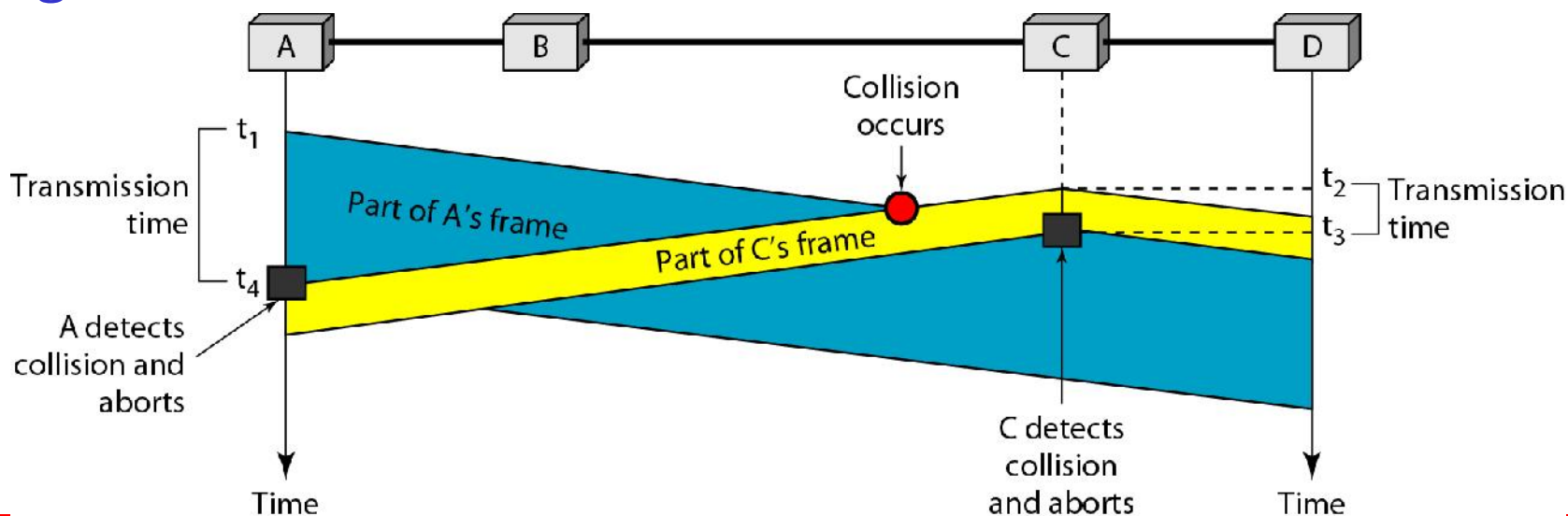
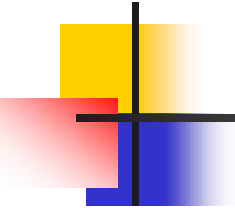


Figure 12.13 *Collision and abortion in CSMA/CD*





Example 12.5 (Minimum frame size)

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is $25.6 \mu\text{s}$, what is the minimum size of the frame?

Solution

The frame transmission time is $T_{fr} = 2 \times T_p = 51.2 \mu\text{s}$. This means, in the worst case, a station needs to transmit for a period of $51.2 \mu\text{s}$ to detect the collision. The minimum size of the frame is $10 \text{ Mbps} \times 51.2 \mu\text{s} = 512 \text{ bits}$ or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet.

Figure 12.14 *Flow diagram for the CSMA/CD (Page 375)*

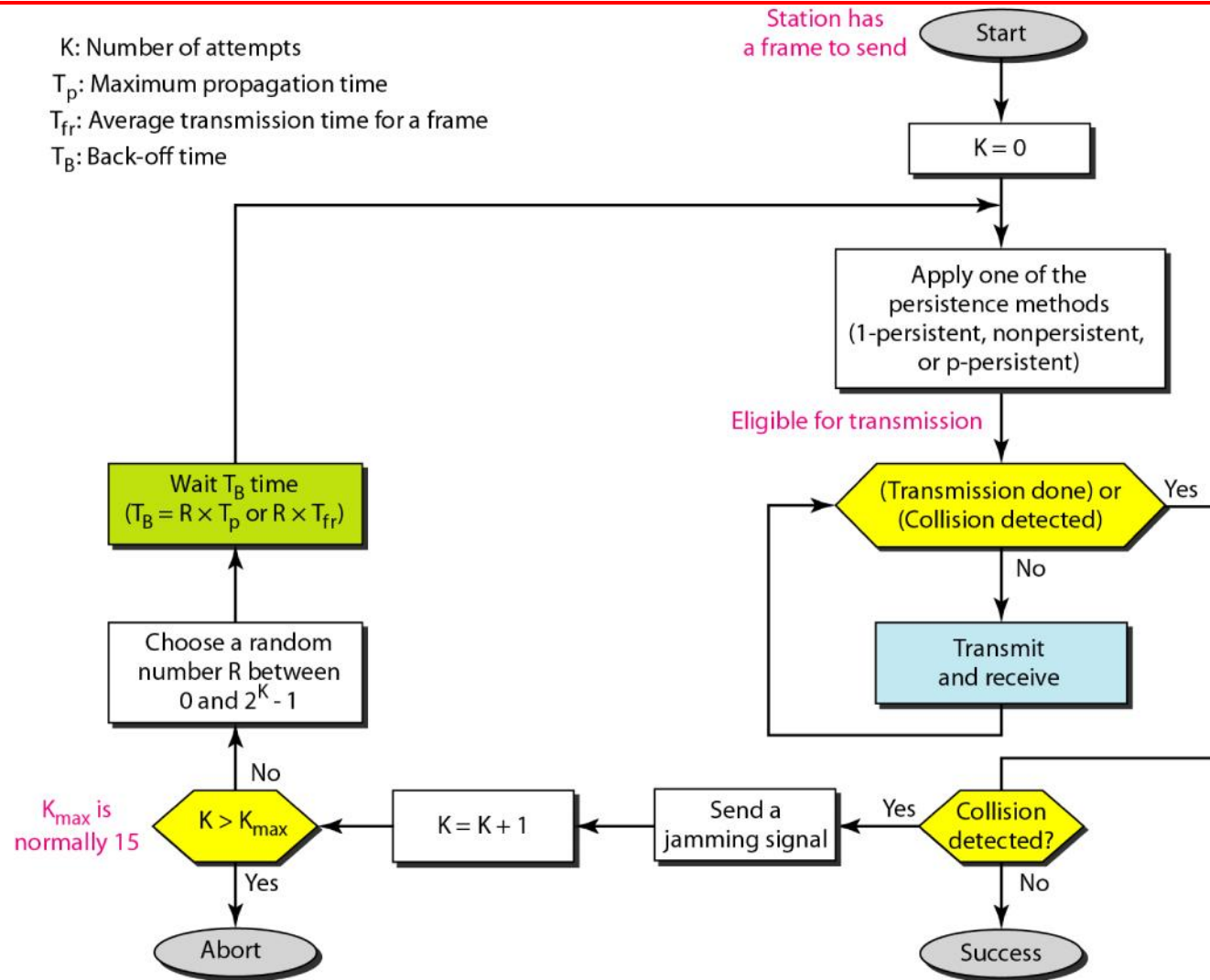
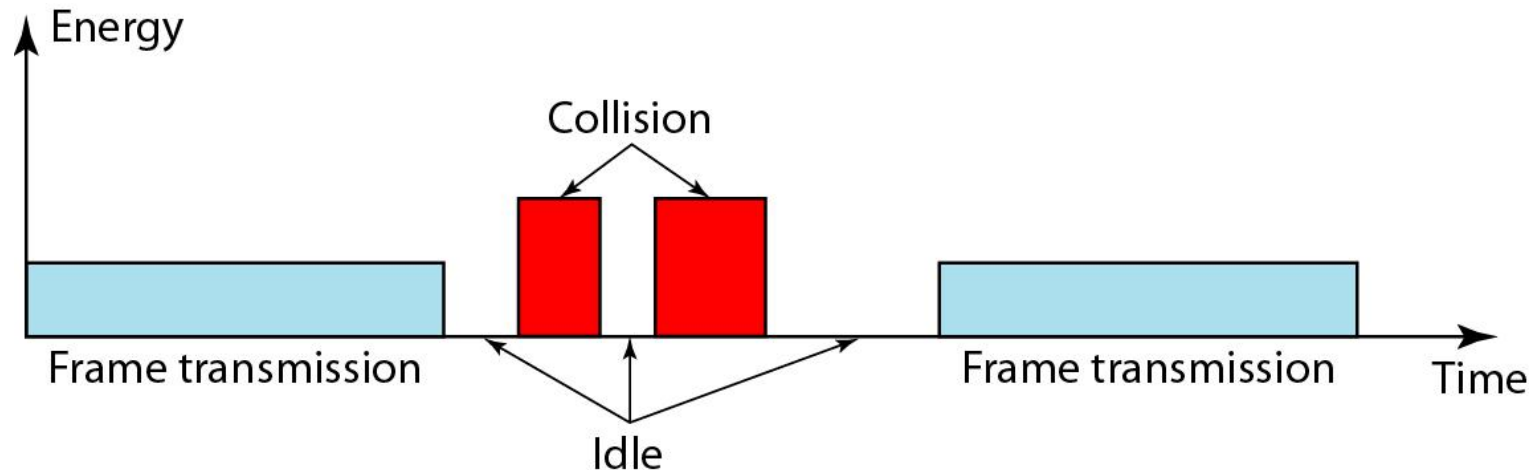


Figure 12.15 *Energy level during transmission, idleness, or collision*



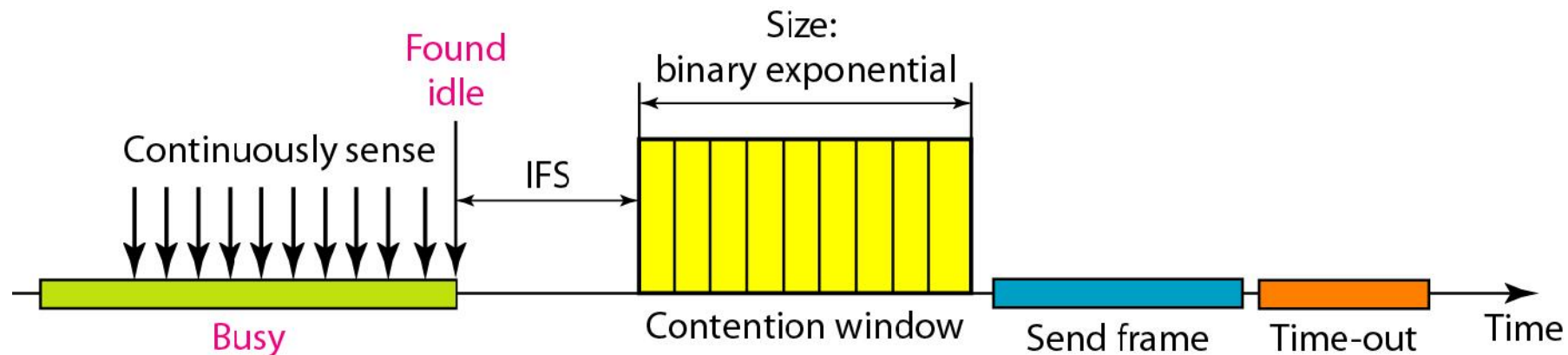
Zero level—channel is idle

Normal level—successfully captured channel and sending frame

Abnormal level—collision and energy twice the normal level

Figure 12.16 *Timing in CSMA/CA*

Avoid collisions on wireless network because they cannot be detected

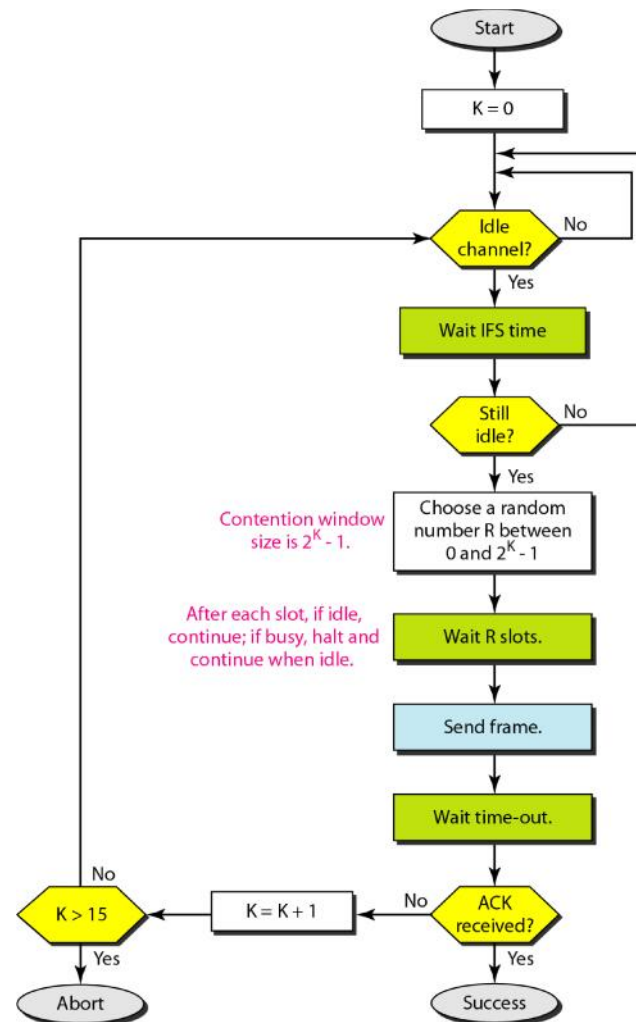


IFS- In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

Contention window- In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

Acknowledgment- Positive acknowledgment and time out timer guarantee receiver has received the frame

Figure 12.17 *Flow diagram for CSMA/CA*



12-2 CONTROLLED ACCESS

*In **controlled access**, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.*

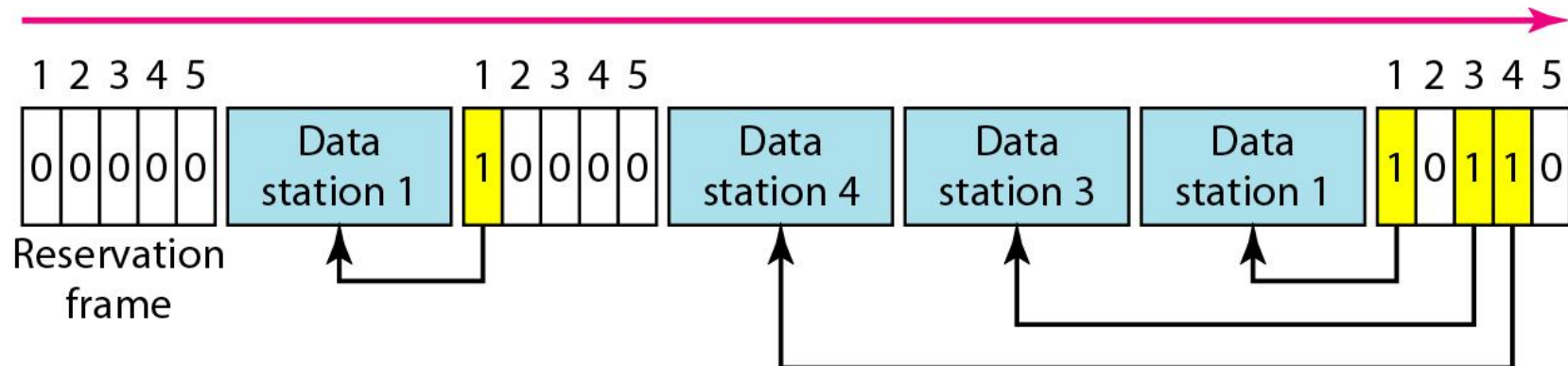
Topics discussed in this section:

Reservation

Polling

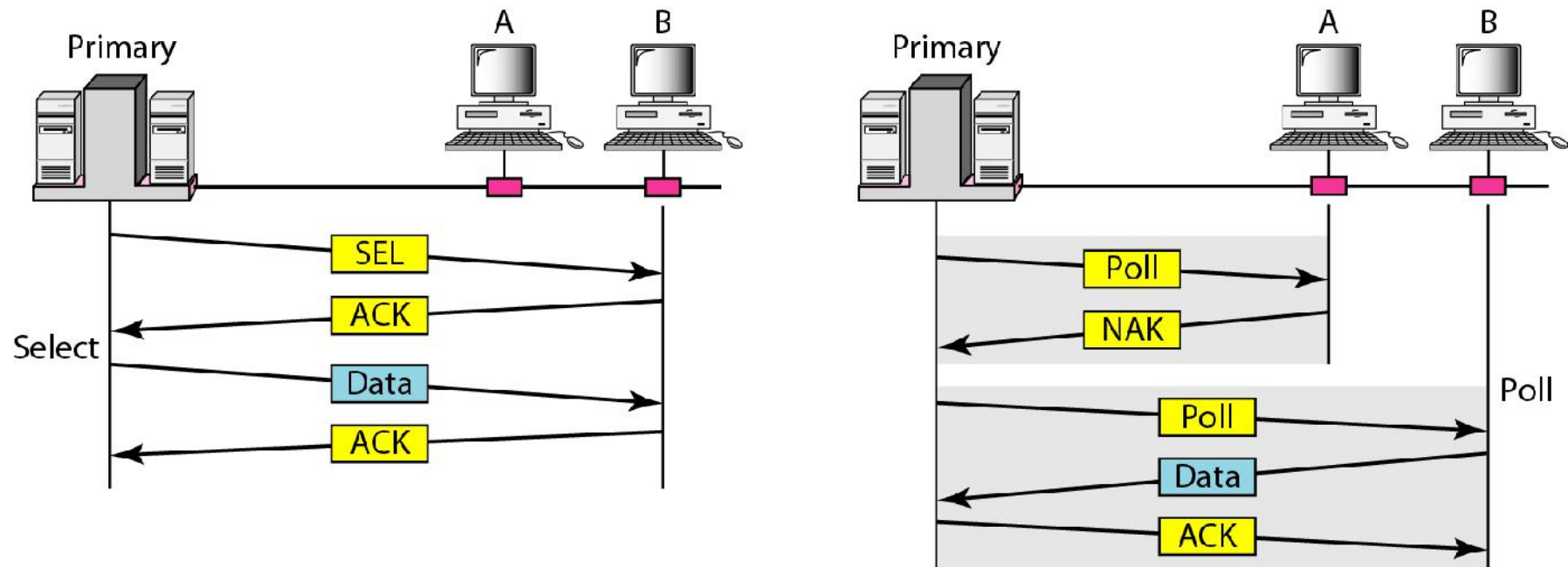
Token Passing

Figure 12.18 *Reservation access method*



Reservation-station needs to make a reservation before sending data

Figure 12.19 *Select and poll functions in polling access method*



Polling – one device as primary station and the other device as secondary station
Select – primary device wants to send data to secondary device, secondary device gets ready to receive
Poll – primary device solicits (ask) transmissions from secondary devices

Token passing – stations in network organized in a logical ring – predecessor and successor

Token – gives station right to access the channel; needs token management

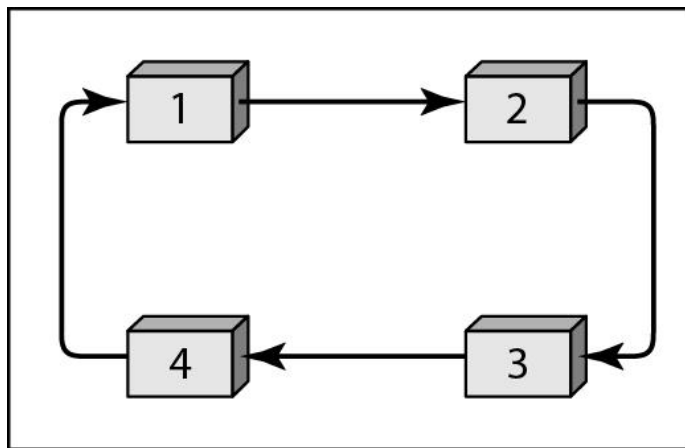
Physical ring – station sends the token to successor

Dual ring – uses second ring which operates in reverse direction

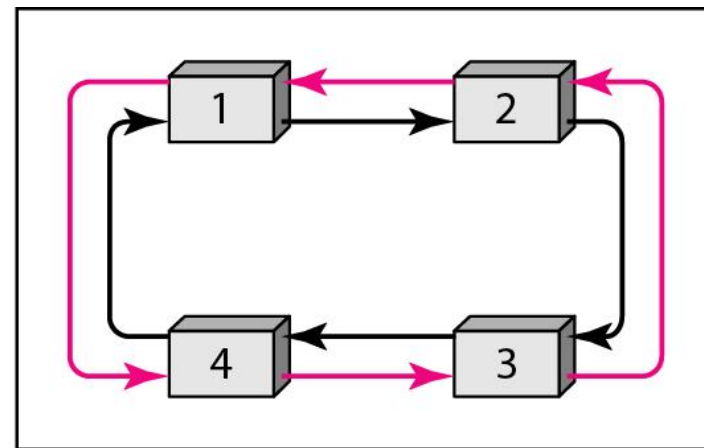
Bus ring (token bus) - stations are connected to single cable called bus, but make logical ring

Star ring - physical topology star, wiring inside hub makes the ring

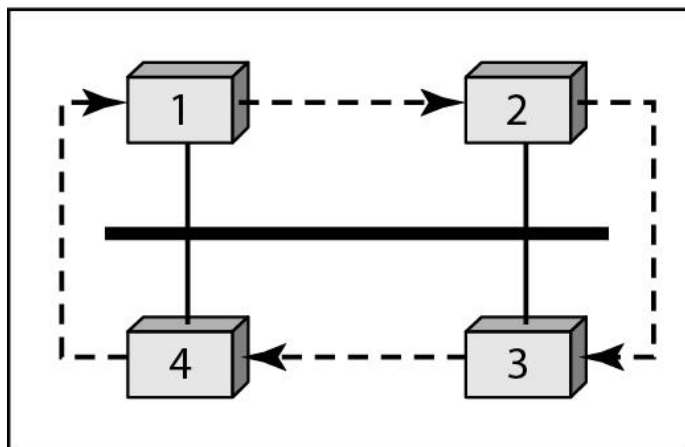
Figure 12.20 *Logical ring and physical topology in token-passing access method*



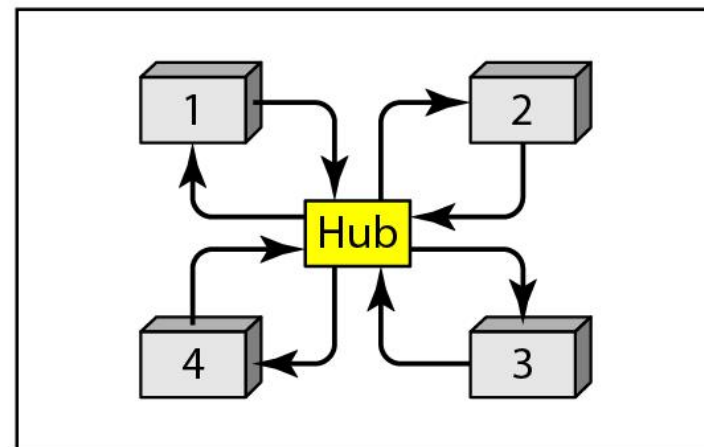
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring